Policy Research Working Paper 11067

Rapid Economic Growth but Rising Poverty Segregation

Will Viet Nam Meet the SDGs for Equitable Development?

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February 2025

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Abstract

Viet Nam is widely regarded as a success story for its impressive economic growth and poverty reduction in the last few decades. Yet, recent evidence indicates that the country's economic growth has not been uniform. Compiling and analyzing new, extensive province-level data from the Vietnam Household Living Standards Surveys spanning 2002 to 2020 and other data sources, this paper finds within-province inequality to be much larger than between-province inequality. Furthermore, this inequality gap has been rising over time. Despite the country's fast poverty reduction, the poor were increasingly segregated in certain provinces, particularly those with a larger ethnic minority population. The analysis finds a beneficial impact of economic growth on poverty reduction, but this can depend on inequality levels. It also finds that greater inequality has had negative effects on economic growth but varying negative effects on different poverty indicators. The paper provides supportive evidence of the beneficial impact of economic transitions from agriculture to non-agriculture. The results suggest that policy makers in Viet Nam should focus on reducing spatial disparities and income inequality to attain sustainable economic development.

This paper is a product of the Development Data Group, Development Economics and the Poverty and Equity Global Department. It is part of a larger effort by the World Bank to provide open access to its research and make a contribution to development policy discussions around the world. Policy Research Working Papers are also posted on the Web at http://www.worldbank.org/prwp. The authors may be contacted at hdang@worldbank.org and vietcuong@vnu.edu.vn.

Rapid Economic Growth but Rising Poverty Segregation:
Will Viet Nam Meet the SDGs for Equitable Development?

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JEL: C15, D31, I31, O10, O57
Keywords: poverty, inequality, pro-poor growth, convergence, household surveys, Viet Nam

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1. Introduction

Poverty and inequality provide two closely related, but different, measures of household welfare. As rising global living standards have led to less poverty, increasingly more attention has been placed on whether the fruits of economic growth are equally distributed to different population groups in a society. The Sustainable Development Goals (SDGs) adopted by the United Nations (UN) offer a most notable example where the needs to reduce both poverty (SDG number 1) and inequality (SDG number 10) are emphasized by the international community. Indeed, a country with good economic growth but a (highly) unequal income distribution may neither be able to shrink its poor nor narrow the undesirable gaps between its population segments.

Viet Nam is widely regarded as a success story for its impressive economic growth and poverty reduction in the last few decades. Its growth has been found to be pro-poor since the early 1990s (Glewwe and Dang, 2011; Nguyen and Pham, 2018; Pimhidzai and Niu, 2021). Yet, recent evidence indicates that the country's economic growth was not uniform: while inequality gaps between urban and rural areas have been found to narrow over time, they have widened within urban and rural areas (Bui and Imai, 2019). Furthermore, poverty rates became concentrated spatially (Lanjouw, Marra, and Nguyen, 2017) and among ethnic minority groups (Benjamin, Brandt, and McCaig, 2017), suggesting that attention can be focused on these particular groups for more effective poverty amelioration.

In this paper, we aim to provide a long-term (and mostly descriptive) review of trends in poverty, inequality, and economic growth in Viet Nam. We contribute to the existing literature in several ways. First, we offer a broad but early assessment of the intertwined relationship between poverty, inequality, and economic growth for the country over the past 20 years. Recent review studies suggest that there is an intricate and complex relationship between these three development outcomes, which require further evidence for better understanding (Cerra *et al.,* 2022; Ferreira *et al.*, 2022). In fact, we offer the first study to investigate all three development outcomes of poverty, inequality, and economic growth for Viet Nam.

Specifically, we investigate several research questions that are highly relevant to policy. Have average incomes across provinces converged (or diverged) over time? If yes, how were the poor segregated? Why were the poor spatially concentrated in some provinces and not others? Did factors such as inequality, urbanization, investment, and volume of government spending play a role in determining poverty reduction? Furthermore, what could we tell about the relationship between economic growth, inequality, and (different measures of) poverty?

Second, our analysis spans the period 2002-2020, which represents the longest-running period for the country that has been examined in an academic study. We complete this task by analyzing 10 rounds of the nationally representative Vietnam Household Living Standards Surveys (VHLSS). Our findings on the trends of poverty, inequality and economic growth, which are further disaggregated into between-province and within-province and regional variations, offer a nuanced picture of the evolution of these outcomes over time. To our knowledge, these findings are new and not available in previous studies.[1]

Finally, we construct a database of panel data at the province level, which allows us to offer more granular analysis than the urban-rural dichotomy analyzed in previous studies. We further

[1] Among the three development outcomes of poverty, inequality, and economic growth that we analyze, existing studies focus on either a single outcome or two outcomes. Their analyses are mostly restricted to survey data before 2010. For example, analyzing earlier VHLSS data covering the early 1990s to the early 2000s, Le and Booth (2014) and Nguyen *et al.* (2007) found widening urban-rural gaps in inequality. Benjamin *et al.* (2017) examine household income inequality during 2002-2014 through several dimensions, including urban-rural, ethnicity, and work sectors. Bui and Imai (2019) examine urban-rural gaps in Viet Nam during 2008-2012 and find education and remittances to be important factors. For other studies that focus on the ethnic gaps in living standards, see Dang (2012) and Fujii (2018).

supplement this database with data on government spending that we collect from the central government and different provincial government websites. As a result, we can zoom in on provinces that are good (or bad) performers. Furthermore, this rich panel database also allows us to employ rigorous econometric modeling techniques to investigate the channels affecting the province-level income distributions.

Our estimation results suggest that within-province inequality has steadily increased over time. Within-province inequality is also much larger than between-province inequality, with the former type of inequality being almost three times the latter type of inequality in 2020. Average incomes and expenditures appear to converge across provinces, while poverty significantly declines during this period. The remaining poor seem to be regionally segregated among provinces with a greater share of ethnic minority population. We find beneficial impact of economic growth on poverty reduction. While the relationship of inequality with headcount poverty appears weak and not statistically significant, inequality is positively and strongly statistically correlated with poverty gap and poverty severity and might impede economic growth. Provinces with greater population density and a larger share of urban population have faster growth, whereas the opposite result holds for those with a greater share of ethnic minority groups. The transitions from an agriculture-based economy toward wage-based and services-based economy could be beneficial for the country's economic growth and poverty reduction.

This paper consists of four sections. In the next section, we describe the data (Section 2.1), discuss the overall trends in economic growth, inequality, and poverty at the country level (Section 2.2) and at the regional level as well as analyze poverty segregation across provinces (Section 2.3). We subsequently investigate in Section 3 the relationship between economic growth, inequality, and poverty, including convergence in inequality. We finally conclude in Section 4.

2. Trends in economic growth, inequality, and poverty

2.1. Data

We compile data from the Vietnam Household Living Standards Surveys (VHLSSs), which has been widely employed by the government, the international community, and academic researchers for poverty and inequality analysis for the country. The VHLSSs have been conducted by the General Statistics Office of Vietnam with technical support from the World Bank every two years since 2002. We compile data on all 58 provinces and five centrally controlled municipalities and supplement this data with other data that we collected.[2] In particular, provincial government spending data for the period 2018-2020 is currently unavailable for all the 63 provinces in any official document. To get the most updated data, we manually collected the state spending and investment spending data for 2018-2020 from several sources including the Ministry of Finance's website, provincial finance departments' websites, and relevant official documents.

The VHLSSs contain detailed data on individuals and households. Household-level data are collected on durables, assets, production, income, and participation in government programs. Individual-level data are collected on demographics, education, employment, health, and migration. The 1999 Population and Housing Census was used as the sampling frame of the VHLSSs during 2002-2008, while the 2009 and 2019 Population and Housing Censuses were used as the sampling frame of the VHLSSs respectively for 2010-2016 and 2018-2020. Around 3,100 communes were chosen as the primary sampling units out of the list of 10,000 communes for the

[2] These municipalities are Can Tho, Da Nang, Hai Phong, Hanoi, and Ho Chi Minh City.

whole country. A village was randomly selected from each commune, and about 15 households were selected randomly from the village.

The large-sample VHLSSs have a sample size of about 46,000 households that are designed to be representative at the provincial level. While these large-sample surveys only collect income data, a sub-sample of the VHLSSs (the small-sample VHLSSs of around 9,000 households) collect data on both income and expenditure. Since policy discourse in the country is typically based on poverty and inequality analysis using expenditure data, we mostly analyze such data in this paper. We obtain the province-level expenditure using Elbers *et al.*'s (2003) small area (poverty-map) estimation method (see Appendix B for more detailed description for the data). For robustness checks, we also provide some alternate estimates using per capita income in Appendix A, which offer qualitatively similar results.

2.2. Overall trends in economic growth, inequality, and poverty

We provide in Table 1 the country trends in (real) per capita income, expenditure, poverty, and inequality. Between 2002 and 2020, per capita incomes increased significantly from 4,565 thousand (Vietnamese) dong to 15,156 thousand dong.[3] Similarly, real per capita expenditure more than tripled from 3,476 thousand dong in 2002 to 14,251 thousand dong in 2020.[4] This rapid economic growth has been widely attributed to important policy changes that took place in Viet Nam in the last two decades (see, e.g., Justino and Litchfield, 2014, Benjamin *et al.*, 2017 for a

[3] On average, slightly more than 18,000 Vietnamese dong equals one US dollar in this period (World Bank, 2018a).
[4] There is a large difference in income and expenditure values between the 2008 and the 2010 VHLSS (Table 1). This difference is mainly because of a change in the recall period in the questionnaires. From 2002 to 2008, the survey asked for household expenditure or income in the past 12 months. However, from 2010 onward, these values were asked for the past month and then multiplied by 12 to estimate annual values. As a result, there is a break in values in per capita expenditure between values before 2010 and those since 2010. We have not made any adjustment to our data following a change in the recall period; for more related discussion on this topic, see Deaton and Kozel (2005).

review). Specifically, the "*Doi Moi*" (renovation) policies introduced in 1986 helped transform Viet Nam from a centrally planned economy to an open, export-oriented economy with high volume of trade and foreign direct investment. In 2001, the United States and Viet Nam signed a Bilateral Trade Agreement, in which the United States granted Viet Nam the status of the Most Favored Nations. As a result, tariffs on Vietnamese exports to the U.S. reduced significantly, triggering export-led economic growth.

Figure 1 plots the distribution of log of per capita expenditure over time. Between 2004 and 2020, the distribution shifted significantly to the right, indicating a rise in average per capita expenditures.[5] However there is not a marked decrease in the variance of the distribution, indicating that inequality did not change rapidly during this period. We estimate three different measures of inequality, namely the Gini index and the Theil L and T indices and show the results in Table 1. All the three measures show steady levels of inequality in per capita expenditure. The average value of the Gini index was 0.37 and close to the lower side of the typical range of 0.3-0.5 for Gini values for per capita expenditures in developing countries (World Bank, 2005).

The Gini index is derived from the Lorenz curve and cannot be written as the sum of a term summarizing within-group inequality and a term summarizing between-group inequality (Bourguignon, 1979). Unlike the Gini index, the Theil index is a generalized entropy (GE) measure, and it can be decomposed into within and between components. Table 2 shows the decomposition of the Theil L index (GE 0; also known as the mean log deviation) and the Theil T index (GE 1). We find that within-province inequality increased over time. Within-province inequality explained about 66% of total inequality in 2002, but more than 70% in 2020. This

[5] The distribution for 2010 (not shown, to make Figure 1 less cluttered) also shifts to the right of that of 2008, which further supports the increase in living standards over time. This also helps lessen potential concerns about comparability issues with the changes in the questionnaires as discussed above.

translates into within-province inequality increasing from about twice higher than between-province inequality in the early 2000s, to about three times higher than between-province inequality in the late 2010s. Thus, within-province inequality has become much more significant over time than between-province inequality.[6]

There were significant differences in inequality levels within the provinces (Appendix A, Table A.1). Compared with the national average of 0.37, the Gini index was greater than 0.40 in Lao Cai, Dien Bien, and Lai Chau in the northern mountain region. In 2020, these provinces still had some of the remarkably high poverty rates (Lao Cai: 21.5%, Dien Bien 46% and Lai Chau: 36%). Inequality and poverty levels were similarly high in the central highlands region (e.g., Kom Tum and Gia Lai had Gini values of 0.41 and 0.39 and poverty rates of 17% and 27% respectively). On the other hand, inequality was lower in Mekong River delta with a Gini index of about 0.31 in Long An and Tien Giang (where poverty rates hover around 1%) and 0.28 in Hung Yen and Thai Binh in the Red river delta (where poverty rates range around 1%).

Finally, we also show in Table 1 the estimates of three Foster-Greer-Thorbecke *(*FGT) indices of poverty (Foster *et al.*, 1984). The poverty rate (i.e., the headcount ratio) measures the incidence or the proportion of the poor in the population, the poverty gap measures the depth of poverty (i.e., the average income shortfall of the poor), whereas the poverty severity index (i.e., the squared poverty gap) takes into account inequality of the income distribution among the poor. All three measures are estimated using the national (expenditure-based) poverty line as well as the World Bank's PPP $3.1 per day poverty line. In tandem with the rapid economic growth, we find that poverty rates declined significantly. Nationwide, the headcount poverty rate decreased from 29%

[6] Interestingly, the world has witnessed rising within-country inequality in the past two decades as well, although within-country inequality still tends to be lower than between-country inequality (Gradin, 2021).

in 2002 to less than 10% and 5% in 2016 and 2020, respectively. Notably, the first goal of the United Nations' Millennium Development Goals was to reduce extreme poverty rates by half between 1990 and 2015 but Viet Nam appears to have well exceeded the target.[7] More summary statistics are provided in Appendix A, Table A.2.

2.3. Regional distribution of poverty

The country-level trends in economic growth, poverty and inequality do not reflect the regional variation in these indicators. In Table A.1 in the Appendix, we present the estimates for each of these indicators in 2020, for all 63 provinces. In the last two decades, although overall poverty declined rapidly in Viet Nam, poverty rates varied significantly across provinces. Poverty rates were lowest in Ho Chi Minh City (1.8% average over time) and neighboring Binh Duong province (3.2% average). Ho Chi Minh City is the largest city and the prime economic center in Viet Nam. The city has numerous export processing zones, industrial parks, colleges, and universities, as well as the largest international airport in the country. After Ho Chi Minh City, Binh Duong is the second highest recipient of foreign direct investment. Both Ho Chi Minh City and Binh Duong province are in the southeast region. Poverty was also lower in the Red River Delta, for instance in the capital city of Hanoi (5.4% average) and the port city of Hai Phong (5% average).[8] On the other hand, poverty rates were very high in northwest provinces of Son La (50% average), Dien Bien (62% average), Lai Chau (60% average) and Ha Giang (56% average) in the northeast. These

[7] The poverty and inequality estimates in Table 1 can differ from those published by the World Bank because of different data sources used. The country's performance is even more impressive if we consider that the poverty rate was 58% in 1992-1993 (World Bank, 1999).

[8] These results concur with those in earlier studies. In particular, McCaig (2011) finds that provinces that were more exposed to the U.S. tariff cuts experienced faster decreases in poverty in the early 2000s. Pham and Mukhopadhaya (2018) also found that poverty rates were lower in the regions of Southeast and Red River Delta.

provinces lie in the inland, mountainous regions, bordering China and the Lao People's Democratic Republic and have more than 80% of their population residing in rural areas. Furthermore, more than 70% of their population consists of ethnic minorities.

Given the large variance in poverty levels across provinces, there is evidence suggesting that poverty has become spatially concentrated over time (Lanjouw *et al.*, 2017). We measure disparity in the regional distribution of poverty by estimating a poverty segregation curve (Dhongde, 2017).[9] The poverty segregation curve is a useful graphical tool to analyze how the regional distribution of the poor changed over time. The curve compares a province's share of the poor population with its share of the overall population. The poor are segregated when provinces' share of the poor does not resemble their share in the overall population. Perfect integration (zero segregation) implies that each province has the same share in the poor and the overall population (poor and non-poor combined). Figure 2 plots the poverty segregation curve for 2002 and 2020. The diagonal line of equality shows that there is zero segregation of the poor. The 2002 curve lies above the 2020 curve and hence dominates the 2020 curve. In other words, there was unambiguous increase in the segregation of the poor in Viet Nam.

Poverty segregation curves may often intersect or overlap, and thus fail to provide a complete rank ordering of inequality. In Table 3, we calculate two indices of segregation, namely the Dissimilarity index and the Gini index. The Dissimilarity index is equal to one-half the sum of the absolute difference between the proportion of the poor and the proportion of the population across provinces. The Gini index is equal to twice the area between the segregation curve and the diagonal

[9] Also see Massey (2016) for a more general discussion on the segregation curve.

of equality.[10] Between 2002 and 2010, there was not a marked increase in segregation. However, since 2012, there was a steady rise in the spatial inequality in the distribution of the poor. Provinces with a cumulative share of about 50% of the total population had about 30% of the poor population in 2002, whereas only about 3% of the poor population in 2020. Over the years, poor provinces such as Dien Bien and Lai Chau saw their share of the poor population increase disproportionately. Despite a rapid decline in average poverty levels, we find that the remaining poor were increasingly segregated in certain provinces in the country.

3. Relationship between economic growth, inequality and poverty

3.1. Factors affecting provincial poverty

We further analyze the VHLSS data to find which factors were highly correlated with provincial poverty levels. Following the existing literature that examines the effects of income and inequality on poverty (Bourguignon, 2003; Kraay, 2006; Chen and Ravallion, 2010; Ferreira, 2010; Marrero *et al.*, 2024), we estimate the following province fixed effects (FE) model

$$P_{i,t} = \alpha + \beta_1 Log(Y_{it}) + \beta_2 G_{it} + \delta X'_{it} + \tau_t + u_i + \varepsilon_{it} \tag{1}$$

In equation (1), $P_{i,t}$ is a poverty index of province i in year t, $Y_{i,t}$ is per capita expenditure, G_{it} is the Gini index, X_{it} is a vector of explanatory variables including high school completion rates, population density (in logarithmic form), shares of the urban population and the ethnic minority population, share of the population with a high school diploma, and different types of investments such as state spending and investment spending. These are the control variables that are widely

[10] The Gini index satisfies the properties of symmetry, scale invariance, and the regressive transfer principle, and provides a consistent ranking of the distributions whenever the segregation curves do not intersect. The dissimilarity index does not satisfy the principle of regressive transfer. See Dhongde (2017) for a detailed discussion on these properties.

used in previous studies. We also include in X_{it} the shares of the population with wage income and non-farm income to control for local economic development. The year dummy variables τ_t and the province dummy variables u_i help capture unobserved trends occurring for the same years and provinces. ε_{it} is the error term. The two coefficients β_1 and β_2 that measure the correlation between per capita expenditure and inequality are of primary interest.[11] The summary statistics are provided in Appendix A, Table A.2.

For better comparison, we use three model specifications for equation (1) where we sequentially add the explanatory variables to the two core variables (log of) per capita expenditure and Gini index. For the first model, we add the province FE. For the second model, we add to model 1 the year FE. Finally, for model 3, we add all the remaining X_{it} control variables, which is our preferred model for interpretation. Table 4 shows that the estimated results for per capita expenditure vary somewhat between model 1 and model 2, which is perhaps due to unobserved trends occurring in different years. But more importantly, once we control for both the province and year FE, overall, the results remain similar between model 2 and model 3.

In particular, Table 4 shows that, holding all other factors constant, a 1 percent rise in a province's per capita expenditure reduced its poverty rate by 0.26 percentage points (column 3), its poverty gap by 0.10 percentage points (column 6), and its poverty severity index by 0.05 percentage points (column 9). Interestingly, holding fixed the per capita expenditure levels, a 1

[11] We also estimate a variant of equation (1) where we include the explanatory variables in lagged forms using both province FE and GMM models. The results are qualitatively similar (Dang et al., 2023). We further show the estimation results with additional control variables for public infrastructure such as the share of villages with year-round passable roads, the share of villages with a daily market, and the distance from the village to the nearest town in Appendix A, Tables A.3 and A.4. Since these variables are computed from the commune module of the VHLSSs, we do not have complete data for these control variables for the whole period of our study. Yet, these tables show that provinces with more villages that have year-round passable roads could reduce poverty and have faster economic growth.

percent rise in the Gini index led to a reduction in the poverty rate of 0.19 percentage points, but this estimate is not statistically significant. But a similar increase in the Gini index has stronger and statistically significant effects on both the depth and severity of poverty (columns 6 and 9). These results generally concur with findings for other countries, which suggest that while economic growth can be beneficial for poverty reduction, this relationship can change depending on inequality levels (Cerra *et al.*, 2022; Ferreira *et al.*, 2022). But our results shed new light on the intricate relationship between poverty versus inequality and growth using the Viet Nam context and further highlight that we should consider other measures of poverty beyond headcount poverty for a more nuanced understanding of these dynamics.

For the other control variables, Table 4 shows that keeping other factors fixed, provinces with a larger share of urban population or ethnic minority population had greater poverty and more poverty depth and poverty severity. Despite being only marginally statistically significant at the 10 percent level, the positive sign of the estimated coefficient of state spending on all the three poverty indexes is a puzzle (columns 3, 6, and 9). A possible explanation is that state spending could have been mainly reserved for improving infrastructure, and poorer (and more remote) provinces were likely to have been allocated more investment. While population density could raise the poverty rate, higher shares of wage or non-farm income could significantly lower it. The magnitude of impact is comparable to that of the province's per capita expenditure. A 1 percent increase in a province's share of non-farm income and wage income reduced its poverty rate by 0.27 and 0.31 percentage points, respectively (column 3).

We further examine factors that are related to a province's poverty segregation, as measured by its dissimilarity index. Table A.5 in Appendix A indicates that richer provinces or provinces with a lower share of urban population tend to have less poverty segregation, while the opposite holds for provinces with more population density, or provinces that have a higher share of wage or non-farm income. These results hold whether or not we control for the Gini index, the other measure of poverty segregation (columns 3 and 4).

3.2. Factors affecting provincial economic growth

In the previous section, we found a strong negative relation between per capita expenditure levels and poverty and a positive relation between inequality and poverty. In this section, we analyze how poverty and inequality in turn affect economic growth. There is an extensive literature on the impact of inequality and poverty on economic growth, but there appears to be inconclusive evidence on the impact (see Baselgia and Foellmi (2022), Cerra *et al.* (2022), and Ferreira *et al.* (2022) for recent reviews).

To analyze the inequality-growth and the poverty-growth links, Marrero and Serven (2022) employ the overlapping-generations model with learning-by-doing and knowledge spillovers (Aghion *et al.*, 1999). In this model, poor people have initial endowment below a minimum consumption level. The poor do not save and do not contribute to the aggregate economic growth. In this setting, this study shows that aggregate income growth depends on the share of people below the poverty threshold (i.e., poverty) and on the distribution of endowments (i.e., inequality). Using VHLSSs data, we estimate their reduced form empirical model:

$$\Delta Y_{i,t} = \pi + \gamma_1 Log\left(Y_{i,t-1}\right) + \gamma_2 G_{i,t-1} + \gamma_3 P_{i,t-1} + \varphi X'_{i,t-1} + \tau_t + u_i + v_{i,t} \qquad (2)$$

where $\Delta Y_{i,t} = Log\left(Y_{i,t}\right) - Log\left(Y_{i,t-1}\right)$.

Note that equation (2) is a combination of the standard empirical poverty and growth regression (Ravallion, 2012) and the standard inequality-growth regression (Forbes, 2000; Berg *et al.*, 2018), which allows for testing the impact of poverty and inequality on expenditure growth in the same equation. As Marrero and Serven (2022) point out, the identifying assumption for the main coefficients of interest γ's in this equation is that poverty is not an almost exact linear combination of expenditure and inequality.

Equation (2) is also a standard model used to test conditional β-convergence in per capita expenditure (Barro and Sala-i-Martin, 1991).[12] Growth of per capita expenditure, which equals the difference between log of current per capita expenditure and log of lagged per capita expenditure, is regressed on lag of the control variables. This model is now modified by adding lagged values of inequality and poverty to the set of growth determinants. Clearly, we should be cautious and interpret the lagged Gini and poverty rate in equation (2) as having a correlational—rather than causal—relationship with income growth. Put differently, compared to equation (1), equation (2) presents a related, but different, hypothesis on the intertwined relationship between income growth, inequality, and poverty where the outcome is a change variable (i.e., growth of per capita expenditure) instead of a level variable (i.e., poverty indicators) as with equation (1).

We estimate equation (2) using both province FE and GMM models, with the GMM model being our preferred model for interpretation. The province FE model can be biased if the log of lagged per capita expenditure is correlated with unobserved variables (but these can serve as useful robustness checks). We address this omitted variable bias as follows. Firstly, we also estimate the model of first-differenced variables, and the first difference transformation removes the time-invariant unobserved effect (u_i). The Arellano–Bond tests for zero autocorrelation of the first order

[12] Figure A.1 in Appendix A shows a scatter plot with initial expenditure levels and growth in expenditure in the following years. The scatter and the fitted line indicate that a negative relation between initial per capita expenditure levels and growth rates, suggesting β-convergence. Moreover, the correlation is higher for the long-term growth. It implies that in the long run provinces tend to be convergent in economic growth.

and second order in first-differenced errors are also reported, and these test statistics for autocorrelation present no evidence of model misspecification (i.e., the AR1 and AR2 test values of -6.41 and 4.25 correspond to p-values of less than 0.01; Table 5, column 6). Secondly, we apply the GMM estimator which was developed by Holtz-Eakin, Newey, and Rosen (1988) and Arellano and Bond (1991). The GMM-type instruments for the log of lagged per capita expenditure are higher order lags of the per capita expenditure variables. Although the exogeneity of these instruments may be questionable, we can perform the overidentification test to assess the validation of the instruments. The Sargan test statistics of overidentifying restrictions is performed and reported at the bottom of Table 5 (column 6). The null hypothesis that overidentifying restrictions are valid is not rejected in all the regressions.

We use the same vector of explanatory variables X_{it} as with equation (1). We also estimate three model specifications, where we sequentially add the explanatory variables to the two core variables: lagged (log of) per capita expenditure and Gini index. That is, we add to the first model specification the province FE, and we add to the second model specification the year FE. Finally, for the third model specification, we add all the remaining X_{it} control variables, which is our preferred model for interpretation. We present the estimation results for equation (2) in Table 5. Similar to Table 4, the second model specification with both province and year FE provides intermediate estimation results between the other models: the absolute values of its estimated lagged expenditures (columns 2 and 5) are larger than those using the first model specification with province FE (columns 1 and 4) but smaller than those using our preferred third model specification with all the control variables (columns 3 and 6). We also re-estimate equation (2) using per capita income instead of per capita expenditure and show the estimates in Appendix A, Table A.6.

In almost all the different specifications in Table 5, the estimated coefficient on lagged log of per capita expenditure is negative and significant at the 1% level. Thus, there is evidence of conditional β-convergence of per capita expenditure across provinces. Provinces with higher expenditure experienced a lower growth rate. The absolute magnitude of the estimate of the lagged log of per capita expenditure is larger in models with control variables than in those without control variables. According to our preferred GMM model with control variables (column 6), if per capita expenditure in the current period increased by 1 percent, the growth rate of per capita expenditure in the next period is 0.60% lower. In other words, if per capita expenditure increases by 1 percent in the current period, per capita expenditure in the next period will increase by 0.40% (i.e., one minus 0.60).

In addition to finding evidence on conditional β-convergence across provinces, Table 5 shows strong negative correlation between inequality and growth in per capita expenditures, suggesting that inequality is harmful for economic growth. Specifically, a 1 percent increase in the Gini index in the current period is associated with a 0.96% decrease in per capita expenditure growth rate in the next period (column 6).[13] However, poverty rates are not statistically significantly correlated with growth. However, when we replace the headcount poverty rate with poverty gap and poverty severity indexes, the estimated coefficient on the Gini loses statistical significance; the other two poverty indexes become strongly statistically significant instead (Appendix A, Tables A.7 and A.8, columns 6). Since these two other poverty indexes better account for inequality compared to the

[13] While this impact may appear larger than those in recent cross-country studies, the difference can be due to different inequality levels and modelling options. For example, Berg *et al.* (2018) find that a 10-percentile increase in net Gini decreases growth on average by 0.48 percentage points per year, but the Gini index range in their study is around 0.30-0.70, which is larger than ours at 0.26-0.48. On the other hand, Forbes (2000) finds the opposite result that a ten-point increase in a country's Gini coefficient is correlated with a 1.3 percent increase in average annual growth over the next five years.

headcount poverty rate, this indicates that a combination of poverty and inequality could be harmful for economic growth.

Regarding other control variables, provinces with more population density or larger shares of urban population or wage and non-farm income have faster growth. While the results with the population variables have opposite effects on poverty, the results with wage and non-farm income are consistent with those discussed earlier with Table 4. These suggest that the transitions from an agriculture-based economy toward wage-based and services-based economy could be beneficial for the country's economic growth and poverty reduction. Our result is consistent with a key finding that movement of labor from agriculture to non-agriculture through structural transformation are a key driver behind economic development in recent studies for Viet Nam (Tarp, 2017; Liu *et al.*, 2022), and other countries (Deininger *et al.*, 2022).

As further robustness checks, we modify equation (2) by adding the interaction terms between lagged per capita expenditure (and income) and lagged values of control variables (state spending, investment spending, population density, share of urban population, share of population with high-school diploma and share of ethnic minority) and show the estimation results in Appendix A, Tables A.9 and A.10. We still find strong evidence on conditional β-convergence in both tables. Furthermore, the interaction terms of expenditure and income with poverty is negative and strongly statistically significant in both tables, and the interaction term of income and inequality is negative in Table A.10. These results suggest that, for the same level of economic development, higher levels of poverty and inequality are generally not conducive to economic growth.

4. Discussion and conclusion

In this paper, we provide a comprehensive analysis of poverty and inequality trends in Viet Nam in the past two decades. We compile new, extensive panel data at the province level over the past two decades using the Vietnam Household Living Standards Surveys and other data sources. We find that although average incomes between provinces tended to converge, there was a significant rise in within-province inequality over time. Within-province inequality was three times larger than between-province inequality in 2020. While economic growth helped reduce poverty, greater inequality could negatively affect poverty gap and poverty severity as well as economic growth. Although poverty levels in the country declined significantly, the poor were increasingly segregated in certain provinces. In particular, provinces with a larger share of ethnic minority groups had greater poverty levels. Economic transitions toward wage and service economies could have beneficial impacts on growth and poverty reduction but could also increase poverty segregation.

SDG 1 calls for an end to poverty in all its forms. Certainly, the goal refers to a broader notion of poverty. Admittedly a limitation of our analysis is that we can focus only on monetary (i.e., consumption and income) poverty and do not measure changes in multi-dimensional poverty in Viet Nam. However, goal 1 also emphasizes the need to address poverty in specific underserved geographic areas within each country. To that effect, our analysis examines a new aspect of poverty in the country and reveals a rise in the segregation of its poor. This new finding lends further support to those in the existing studies that poorer population groups are both spatially and ethnically concentrated (Benjamin, Brandt, and McCaig, 2017; Lanjouw, Marra, and Nguyen, 2017; Fujii, 2018). As such, although Viet Nam has made substantial progress in reducing overall poverty, geographically targeted policies can be designed to reach out to the remaining poor.

Another of our contributions is to highlight the importance of reducing inequality. SDG 10 urges policy makers to accelerate progress towards lowering inequality within countries. Although inequality in Viet Nam is lower compared to many other developing countries, we found a rising share of inequality within provinces. Greater inequality levels were negatively correlated with economic growth and positively correlated with poverty levels.

The recent Covid-19 pandemic can offer an illustrative example for both the importance of reducing inequality and poverty segregation. Analyzing Labor Force Surveys data spanning the pandemic, Dang, Nguyen, and Carletto (2023) find that the pandemic had far stronger effects on low-wage workers; specifically, it increased the proportion of below-minimum wage workers by 32% and worsened various wage equality indexes. This study also finds that the pandemic effects were smaller in provinces with greater openness to the global economy (as measured by the share of exports and imports in provinces' GDP). Reviewing the literature that studied the pandemic's effects on vulnerable population groups in Viet Nam, Dang and Do (2023) also find that poor households, informal workers, people in poor health, ethnic minority groups, and other disadvantaged groups have been most affected by Covid-19. These results further highlight that inequality in the country can be deepened during times of crisis, affecting the more vulnerable groups that are most in need of assistance.

Indeed, geographical disadvantages could explain most of the non-farm participation gap in disadvantaged communities (World Bank, 2019). Furthermore, while national target programs that specially support poorer communes have sustained high commune level investments, a smaller share went to the poorest communes (Pimhidzai and Niu, 2021). As such, place-based poverty interventions can help effectively target and mitigate the disadvantages of fewer economic activities in lagging areas, which typically have less population density. Investment in both digital and physical infrastructure, such as building better Internet connections and roads, is beneficial for integrating these communities into the national (and global) economy.

Promising directions for future research include investigation of the impacts of interventions such as place-based poverty programs in Viet Nam as well as experimenting with different analytical frameworks that can provide more insights into the intricate relationship between poverty, inequality, and economic growth.

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Table 1: National trends in per capita expenditure, inequality, and poverty

Indicators	Years									
	2002	2004	2006	2008	2010	2012	2014	2016	2018	2020
Per capita income ('000 D)	4565.9	5335.9	6176.6	6414.9	9305.1	9680.7	10600.6	12044.0	14861.7	15156.3
	(88.1)	(142.6)	(140.5)	(171.4)	(225.8)	(184.7)	(144.5)	(172.9)	(223.3)	(197.1)
Per capita expenditure ('000 D)	3476.1	4009.7	4519.2	4560.0	8520.3	8902.6	9586.5	11577.7	12376.8	14250.9
	(63.0)	(113.1)	(113.9)	(93.6)	(147.5)	(121.9)	(128.9)	(190.7)	(170.4)	(218.8)
Poverty estimate using the national expenditure poverty line										
Poverty rate (%)	28.8	19.5	16.0	14.5	20.7	17.2	13.5	9.8	7	4.7
	(0.6)	(1.0)	(0.8)	(0.8)	(0.6)	(0.6)	(0.5)	(0.5)	(0.4)	(0.3)
Poverty gap index	0.069	0.047	0.038	0.035	0.059	0.045	0.037	0.027	0.02	0.011
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.001)
Poverty severity index	0.024	0.017	0.014	0.012	0.024	0.017	0.015	0.010	0.008	0.004
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Poverty estimate using the poverty line of 3.1$ PPP/day										
Poverty rate (%)	56.2	45.3	34.7	35.7	9.6	6.5	5.7	4.3	3.2	1.4
	(0.8)	(1.7)	(1.5)	(1.4)	(0.4)	(0.4)	(0.3)	(0.4)	(0.3)	(0.2)
Poverty gap index	0.187	0.141	0.103	0.100	0.023	0.014	0.013	0.010	0.007	0.003
	(0.003)	(0.006)	(0.005)	(0.005)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Poverty severity index	0.081	0.061	0.042	0.040	0.008	0.005	0.005	0.003	0.001	0.023
	(0.002)	(0.003)	(0.002)	(0.002)	(0.001)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Gini	0.370	0.370	0.358	0.356	0.393	0.357	0.348	0.381	0.357	0.368
	(0.003)	(0.004)	(0.004)	(0.005)	(0.006)	(0.004)	(0.004)	(0.005)	(0.004)	(0.005)
Theil L index	0.221	0.224	0.212	0.208	0.259	0.213	0.205	0.246	0.219	0.231
	(0.008)	(0.007)	(0.006)	(0.006)	(0.009)	(0.006)	(0.006)	(0.008)	(0.006)	(0.007)
Theil T index	0.249	0.241	0.227	0.227	0.294	0.230	0.216	0.265	0.226	0.247
	(0.010)	(0.007)	(0.008)	(0.008)	(0.014)	(0.009)	(0.009)	(0.013)	(0.007)	(0.011)

Note: i) Authors' estimation from VHLSSs. ii) Standard errors in parentheses. iii) Per capita income and expenditure in thousand VND, adjusted to constant prices in 2002

Table 2: Percent share of inequality in per capita expenditures within and between provinces

Inequality index	Years									
	2002	2004	2006	2008	2010	2012	2014	2016	2018	2020
Theil L										
Within	65.8	68.8	73.8	76.4	72.1	77.9	78.4	72.3	77.4	73.2
Between	34.2	31.2	26.2	23.6	27.9	22.1	21.6	27.7	22.6	26.8
Theil T										
Within	64.6	67.4	73.7	76.2	73.5	78.6	79.1	72.9	78.1	74.1
Between	35.4	32.6	26.3	23.8	26.5	21.4	20.9	27.1	21.9	25.9

Note: i) Authors' estimation from VHLSSs

Table 3: Indices measuring provincial segregation

Year	Dissimilarity Index	Gini Index
2002	0.21	0.30
2004	0.26	0.36
2006	0.28	0.38
2008	0.23	0.32
2010	0.24	0.34
2012	0.28	0.38
2014	0.32	0.45
2016	0.40	0.53
2018	0.50	0.66
2020	0.50	0.61

Note: i) Authors' estimation based on VHLSSs data

Table 4: Factors related to provincial poverty

Explanatory variables	Dependent variable is the poverty indexes of provinces								
	Poverty headcount			Poverty gap index			Poverty severity index		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log of per capita expenditure	-0.420***	-0.328***	-0.261***	-0.036***	-0.104***	-0.099***	-0.013***	-0.047***	-0.047***
	(0.013)	(0.061)	(0.054)	(0.003)	(0.015)	(0.015)	(0.001)	(0.009)	(0.009)
Gini index	-0.507**	-0.350*	-0.125	0.180**	0.085	0.136**	0.086**	0.040	0.066*
	(0.203)	(0.201)	(0.166)	(0.073)	(0.055)	(0.055)	(0.041)	(0.036)	(0.037)
Log of state spending			0.002*			0.001*			0.000*
			(0.001)			(0.000)			(0.000)
Log of investment spending			0.001			0.002			0.002
			(0.005)			(0.002)			(0.001)
Log of population density			0.293***			-0.007			-0.013
			(0.099)			(0.021)			(0.011)
Share of urban population			0.003**			0.001**			0.000**
			(0.001)			(0.000)			(0.000)
Share of population with high-school diploma			0.060			-0.091*			-0.047
			(0.297)			(0.053)			(0.031)
Share of ethnic minority population			0.001***			0.000***			0.000***
			(0.000)			(0.000)			(0.000)
Share of wage income			-0.310***			-0.026			-0.005
			(0.106)			(0.031)			(0.016)
Share of non-farm income			-0.273***			-0.044			-0.019
			(0.102)			(0.027)			(0.015)
Constant	4.078***	3.355***	3.160***	0.311***	0.898***	0.795***	0.112***	0.400***	0.343***
	(0.100)	(0.479)	(0.419)	(0.025)	(0.119)	(0.106)	(0.013)	(0.073)	(0.063)
Year FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	630	630	630	630	630	630	630	630	630
Number of provinces	63	63	63	63	63	63	63	63	63
R^2	0.873	0.892	0.158	0.550	0.716	0.843	0.507	0.647	0.746
σ_u	0.0584	0.0609	0.316	0.0430	0.0335	0.0218	0.0217	0.0173	0.0129
σ_e	0.0724	0.0609	0.0545	0.0247	0.0175	0.0167	0.0129	0.0101	0.00968
ρ	0.394	0.501	0.971	0.752	0.786	0.632	0.739	0.746	0.640

Notes: i) Authors' estimation from VHLSSs using province FE models, ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01, ** p<0.05, * p<0.1.

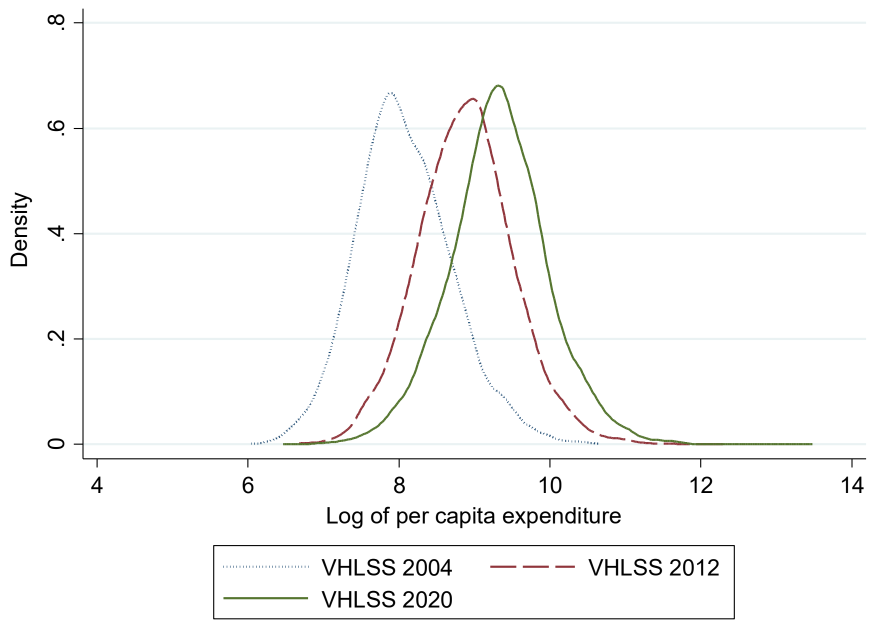
Table 5: Relation between expenditure growth, poverty, and inequality

Explanatory variables	Dependent variable is the growth of per capita expenditure (Log Y_t – Log Y_{t-1})					
	FE			GMM		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged log of per capita expenditure	-0.195***	-0.819***	-0.852***	-0.149***	-0.187***	-0.595***
	(0.033)	(0.058)	(0.062)	(0.044)	(0.049)	(0.091)
Lagged Gini index	-0.631*	0.100	0.078	-0.581***	-0.853***	-0.960***
	(0.362)	(0.146)	(0.156)	(0.203)	(0.161)	(0.166)
Lagged poverty rate	-0.170**	-0.070	-0.069	-0.055	0.055	-0.111
	(0.073)	(0.068)	(0.071)	(0.102)	(0.100)	(0.078)
Lagged log of other State spending			0.002**			0.001
			(0.001)			(0.002)
Lagged log of investment spending			0.001			0.000
			(0.005)			(0.008)
Lagged log of population density			-0.067			0.046***
			(0.071)			(0.014)
Lagged share of urban population			-0.001			0.003***
			(0.001)			(0.001)
Lagged share of population with high-school diploma			0.051			0.060
			(0.283)			(0.171)
Lagged share of ethnic minority population			0.001***			-0.000
			(0.000)			(0.000)
Lagged share of wage income			0.252**			0.373***
			(0.095)			(0.111)
Lagged share of non-farm income			0.149			0.238**
			(0.103)			(0.096)
Constant	2.101***	6.704***	6.732***	1.655***	1.861***	5.082***
	(0.369)	(0.469)	(0.496)	(0.390)	(0.422)	(0.703)
Year FE	No	Yes	Yes			
Province FE	Yes	Yes	Yes			
Observations	567	567	567	567	567	567
R2/ Chi2	0.093	0.894	0.903	2298	3926	7831
Number of provinces	63	63	63	63	63	63

Arellano-Bond test for AR(1)	-7.365	-7.437	-6.411
Arellano-Bond test for AR(2)	2.939	2.837	4.253
Sargan test	311.1	310.8	135

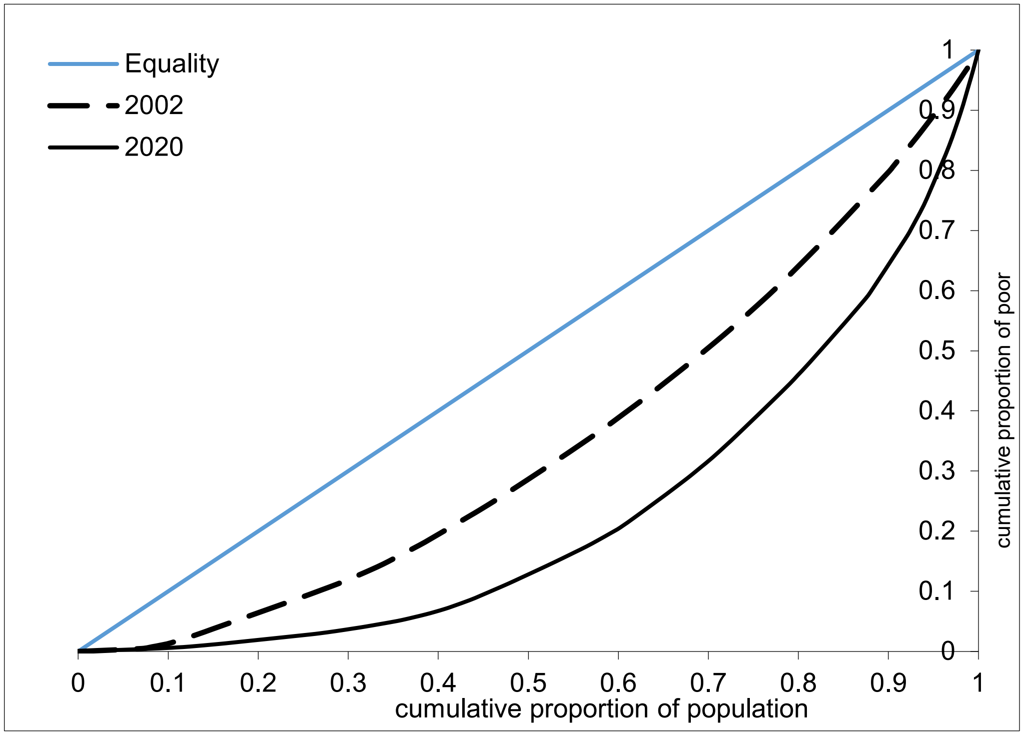
Note: i) Authors' estimation from VHLSSs using province FE model (columns 1 to 3) and GMM model (columns 4 to 6), ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01, ** p<0.05, * p<0.1.

Figure 1: Density of log of per capita expenditure over time



Note: i) Authors' estimation based on VHLSSs data

Figure 2: Segregation of the poor across provinces in Viet Nam: 2002-2020



Note: i) Authors' estimation based on VHLSSs data

Supporting Information for review and online publication only

Appendix A: Additional tables and figures

Table A.1: Average income, inequality, and poverty rates among provinces in Viet Nam in 2020

Province name	Province code	VHLSS 2004				VHLSS 2012				VHLSS 2020			
		Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini	Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini	Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini
Red River Delta													
Hà Nội	1	7,569	6,151	9.0	0.318	34,292	34,666	6.7	0.350	70,183	68,586	0.4	0.321
Quảng Ninh	22	8,061	5,411	10.3	0.342	30,562	27,687	13.7	0.368	53,041	47,940	3.7	0.319
Vĩnh Phúc	26	4,847	3,534	17.8	0.270	22,186	22,806	13.7	0.310	58,206	49,875	0.5	0.300
Bắc Ninh	27	5,891	4,482	8.4	0.279	29,435	26,730	6.0	0.306	68,768	54,120	0.0	0.281
Hải Dương	30	5,414	4,175	11.1	0.266	24,565	23,870	8.3	0.294	57,274	49,202	0.3	0.297
Hải Phòng	31	6,484	5,565	8.1	0.363	30,426	27,745	6.5	0.311	67,333	58,427	0.3	0.302
Hưng Yên	33	5,156	3,551	15.9	0.268	21,704	23,985	8.8	0.290	51,696	44,467	0.8	0.286
Thái Bình	34	4,586	3,676	14.8	0.259	20,488	21,297	12.3	0.295	56,373	44,839	1.1	0.275
Hà Nam	35	4,284	3,431	23.3	0.283	21,282	22,668	13.1	0.303	51,912	49,058	1.1	0.311
Nam Định	36	4,860	3,680	18.1	0.303	21,864	22,558	9.5	0.294	57,699	48,829	0.9	0.313
Ninh Bình	37	4,436	3,813	15.5	0.291	20,385	22,692	11.7	0.314	50,993	47,557	1.0	0.306
Northern Mountain													
Hà Giang	2	2,965	2,395	57.2	0.371	10,131	10,403	70.5	0.381	22,811	22,916	41.1	0.466
Cao Bằng	4	3,344	3,009	43.0	0.350	13,386	14,279	49.2	0.388	27,336	29,094	24.6	0.424
Bắc Kạn	6	3,265	2,730	42.7	0.312	14,306	15,161	42.2	0.356	27,778	28,930	18.7	0.391
Tuyên Quang	8	4,085	3,311	34.4	0.339	14,666	16,586	38.2	0.366	37,344	32,360	8.1	0.317
Lào Cai	10	3,362	2,937	53.5	0.400	13,831	15,817	51.6	0.441	30,663	29,258	21.5	0.426
Điện Biên	11	2,690	2,058	68.8	0.355	10,445	9,985	72.7	0.383	21,853	21,397	45.6	0.458
Lai Châu	12	2,579	1,949	70.9	0.339	9,424	9,619	74.4	0.400	24,449	21,207	35.9	0.385
Sơn La	14	3,326	2,416	59.0	0.372	12,526	12,048	60.4	0.361	27,329	24,156	30.4	0.393
Yên Bái	15	3,935	3,190	36.1	0.343	13,772	16,261	43.9	0.389	32,735	32,504	18.2	0.397
Hoà Bình	17	3,491	2,816	48.9	0.365	14,762	16,561	37.2	0.375	34,479	33,871	9.1	0.385
Thái Nguyên	19	4,761	4,005	23.6	0.340	21,266	22,855	16.5	0.341	43,535	42,082	4.0	0.309
Lạng Sơn	20	4,184	3,217	38.0	0.359	14,451	15,290	46.5	0.392	30,544	30,678	10.4	0.340
Bắc Giang	24	4,709	3,553	20.9	0.301	19,870	19,337	23.4	0.319	50,049	42,191	1.7	0.283
Phú Thọ	25	4,441	3,454	25.9	0.313	18,987	20,477	18.8	0.331	42,208	42,264	2.6	0.315
Central Coast													
Thanh Hoá	38	3,727	3,040	34.9	0.313	15,455	18,394	27.3	0.332	47,451	37,421	4.5	0.331
Nghệ An	40	3,750	3,087	33.2	0.304	16,561	19,018	26.2	0.343	49,245	36,426	8.9	0.348

Province name	Province code	VHLSS 2004				VHLSS 2012				VHLSS 2020			
		Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini	Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini	Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini
Hà Tĩnh	42	3,690	3,218	31.6	0.305	16,259	20,775	17.9	0.318	42,545	38,686	4.7	0.342
Quảng Bình	44	3,617	3,158	33.9	0.308	17,270	20,239	23.3	0.349	43,326	37,591	8.7	0.353
Quảng Trị	45	3,657	2,956	37.5	0.313	16,140	18,386	30.5	0.362	36,003	36,802	12.8	0.391
Thừa Thiên Huế	46	4,578	3,945	22.1	0.340	19,724	22,144	14.7	0.340	41,907	40,520	4.6	0.328
Đà Nẵng	48	8,043	6,859	2.9	0.305	34,565	35,850	3.2	0.336	61,791	70,096	0.3	0.321
Quảng Nam	49	3,924	3,295	28.4	0.300	17,434	18,872	22.5	0.312	44,941	42,968	4.5	0.320
Quảng Ngãi	51	4,048	3,499	22.6	0.292	15,684	18,303	25.7	0.324	40,594	38,232	8.0	0.361
Bình Định	52	5,021	4,107	21.1	0.333	20,673	21,349	14.6	0.313	43,066	44,340	2.1	0.336
Phú Yên	54	4,516	3,627	22.5	0.303	17,413	19,321	16.9	0.297	39,255	38,622	6.4	0.319
Khánh Hoà	56	5,676	4,555	16.5	0.340	21,138	24,903	14.3	0.348	40,098	44,819	3.8	0.318
Ninh Thuận	58	4,667	4,494	20.8	0.374	17,061	20,129	21.9	0.325	36,271	39,373	7.9	0.369
Bình Thuận	60	5,338	4,924	10.9	0.339	21,012	22,304	14.1	0.298	49,947	43,601	2.1	0.310
Central Highlands													
Kon Tum	62	4,084	3,015	47.8	0.402	17,072	16,731	47.6	0.422	36,165	33,221	16.8	0.411
Gia Lai	64	4,431	3,047	46.1	0.381	22,246	17,663	36.3	0.375	29,994	27,393	26.7	0.391
Đắk Lắk	66	4,622	3,211	37.0	0.372	20,877	18,844	30.4	0.365	35,057	33,144	14.1	0.349
Đắk Nông	67	4,257	2,975	32.0	0.295	20,059	18,240	23.9	0.314	35,779	36,117	7.5	0.333
Lâm Đồng	68	5,324	3,813	26.6	0.339	22,353	24,150	18.7	0.345	47,473	43,901	5.2	0.358
Southeast													
Bình Phước	70	5,848	4,502	14.4	0.319	26,671	22,980	11.8	0.291	50,678	48,115	3.4	0.362
Tây Ninh	72	5,721	4,627	11.3	0.312	23,193	21,192	14.0	0.284	53,535	47,888	0.8	0.312
Bình Dương	74	9,335	6,379	3.4	0.307	42,197	27,204	6.2	0.294	84,979	53,861	0.0	0.287
Đồng Nai	75	8,140	5,571	6.8	0.313	29,374	25,884	8.9	0.310	70,667	59,240	0.9	0.319
Bà Rịa - Vũng Tàu	77	7,868	6,372	6.2	0.345	33,363	29,922	5.2	0.318	54,245	68,820	0.2	0.344
Hồ Chí Minh	79	13,943	9,490	1.9	0.307	40,295	34,009	3.7	0.321	77,646	85,497	0.0	0.329
Mekong River Delta													
Long An	80	6,000	4,460	9.7	0.287	23,282	21,779	11.3	0.290	54,063	42,739	1.7	0.308
Tiền Giang	82	5,750	4,399	10.7	0.292	22,983	23,416	10.9	0.303	54,663	41,873	0.6	0.305
Bến Tre	83	4,999	4,121	12.8	0.295	20,071	20,014	15.5	0.305	47,149	37,194	1.7	0.288
Trà Vinh	84	4,755	3,614	20.3	0.301	18,860	17,416	27.8	0.331	50,631	33,305	8.3	0.335
Vĩnh Long	86	5,077	4,032	15.0	0.288	20,972	22,020	14.5	0.321	43,349	39,023	2.0	0.293
Đồng Tháp	87	5,687	4,492	16.7	0.458	19,400	19,866	19.3	0.321	52,289	34,445	4.5	0.289
An Giang	89	6,218	3,883	20.3	0.297	22,574	17,921	20.3	0.292	44,669	33,414	6.5	0.323
Kiên Giang	91	6,130	3,819	22.4	0.331	23,627	18,770	28.1	0.362	62,005	32,432	10.6	0.335
Cần Thơ	92	6,292	4,886	13.4	0.352	28,090	23,662	13.4	0.333	64,735	44,047	1.8	0.326
Hậu Giang	93	5,387	3,486	22.0	0.286	19,450	18,489	27.9	0.333	53,440	35,402	5.3	0.302

Province name	Province code	VHLSS 2004				VHLSS 2012				VHLSS 2020			
		Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini	Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini	Per capita income (thousand VND)	Per capita expend. (thousand VND)	Poverty rate (%)	Gini
Sóc Trăng	94	4,742	3,437	24.7	0.304	17,860	17,636	29.4	0.349	47,319	32,997	7.7	0.360
Bạc Liêu	95	5,610	3,583	22.8	0.305	24,444	19,485	22.8	0.326	51,605	34,829	7.2	0.357
Cà Mau	96	6,176	3,950	19.4	0.318	19,811	18,571	22.3	0.317	39,848	34,263	6.8	0.323

Note: i) Authors' estimation from the 2020 VHLSS.

Table A.2: Summary statistics

Variables	Obs	Mean	Std. Dev.	Min	Max
Headcount poverty	630	0.249	0.252	0.000	0.895
Poverty gap index	630	0.059	0.063	0.000	0.345
Poverty severity index	630	0.024	0.030	0.000	0.188
Log of per capita expenditure	630	8.724	0.563	7.411	10.148
Gini index	630	0.331	0.040	0.257	0.475
Log of state spending	630	15.085	1.761	0.000	18.180
Log of investment spending	630	13.927	1.130	8.397	17.275
Log of population density	630	-1.261	0.999	-3.413	1.482
Share of urban population	630	25.337	16.499	5.023	87.511
Share of population with high-school diploma	630	0.072	0.036	0.006	0.237
Share of ethnic minority population	630	20.851	27.157	0.000	97.066
Share of wage income	630	0.369	0.107	0.145	0.694
Share of non-farm income	630	0.207	0.061	0.045	0.559

Table A.3: Factors related to provincial poverty, with additional control variables

Explanatory variables	Dependent variable is the poverty indexes of provinces								
	Poverty headcount			Poverty gap index			Poverty severity index		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Log of per capita expenditure	-0.420***	-0.328***	-0.274***	-0.036***	-0.104***	-0.103***	-0.013***	-0.047***	-0.049***
	(0.013)	(0.061)	(0.057)	(0.003)	(0.015)	(0.015)	(0.001)	(0.009)	(0.009)
Gini index	-0.507**	-0.350*	-0.090	0.180**	0.085	0.144***	0.086**	0.040	0.069*
	(0.203)	(0.201)	(0.158)	(0.073)	(0.055)	(0.052)	(0.041)	(0.036)	(0.035)
Log of state spending			0.001*			0.000			0.000
			(0.001)			(0.000)			(0.000)
Log of investment spending			0.002			0.003*			0.002**
			(0.005)			(0.002)			(0.001)
Log of population density			0.290***			-0.010			-0.015
			(0.102)			(0.021)			(0.011)
Share of urban population			0.003***			0.001***			0.000***
			(0.001)			(0.000)			(0.000)
Share of population with high-school diploma			0.143			-0.041			-0.017
			(0.299)			(0.047)			(0.027)
Share of ethnic minority population			0.001***			0.000***			0.000***
			(0.000)			(0.000)			(0.000)
Share of wage income			-0.316***			-0.032			-0.009
			(0.104)			(0.031)			(0.017)
Share of non-farm income			-0.258**			-0.042			-0.020
			(0.113)			(0.030)			(0.016)
Share of village with year-round passable road			-0.064			-0.040***			-0.024***
			(0.046)			(0.011)			(0.007)
Share of village with daily market			0.039			0.018			0.010*
			(0.041)			(0.011)			(0.006)
Distance from village to the nearest town			-0.007			-0.001			0.000
			(0.011)			(0.003)			(0.002)
Constant	4.078***	3.355***	3.238***	0.311***	0.898***	0.814***	0.112***	0.400***	0.348***
	(0.100)	(0.479)	(0.440)	(0.025)	(0.119)	(0.099)	(0.013)	(0.073)	(0.056)
Year FE	No	Yes	Yes	No	Yes	Yes	No	Yes	Yes

Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	630	630	628	630	630	628	630	630	628
Number of provinces	63	63	63	63	63	63	63	63	63
R2	0.873	0.892	0.160	0.550	0.716	0.839	0.507	0.647	0.741
σ_u	0.0584	0.0609	0.317	0.0430	0.0335	0.0213	0.0217	0.0173	0.0140
σ_e	0.0724	0.0609	0.0542	0.0247	0.0175	0.0163	0.0129	0.0101	0.00946
ρ	0.394	0.501	0.971	0.752	0.786	0.631	0.739	0.746	0.686

Notes: i) Authors' estimation from VHLSSs using province FE models, ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01, ** p<0.05, * p<0.1.

Table A.4: Relation between expenditure growth, poverty, and inequality, , with additional control variables

Explanatory variables	Dependent variable is the growth of per capita expenditure (Log Yt – Log Yt-1)					
	FE			GMM		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged log of per capita expenditure	-0.195***	-0.819***	-0.863***	-0.149***	-0.187***	-0.816***
	(0.033)	(0.058)	(0.065)	(0.044)	(0.049)	(0.110)
Lagged Gini index	-0.631*	0.100	0.115	-0.581***	-0.853***	-0.233
	(0.362)	(0.146)	(0.160)	(0.203)	(0.161)	(0.305)
Lagged poverty rate	-0.170**	-0.070	-0.073	-0.055	0.055	-0.203
	(0.073)	(0.068)	(0.077)	(0.102)	(0.100)	(0.127)
Lagged log of other State spending			0.002*			-0.001
			(0.001)			(0.002)
Lagged log of investment spending			0.001			-0.012
			(0.005)			(0.012)
Lagged log of population density			-0.064			0.058***
			(0.072)			(0.021)
Lagged share of urban population			-0.001			0.005***
			(0.001)			(0.001)
Lagged share of population with high-school diploma			0.096			-0.388
			(0.291)			(0.386)
Lagged share of ethnic minority population			0.001***			0.001
			(0.000)			(0.000)
Lagged share of wage income			0.244**			-0.050
			(0.096)			(0.192)
Lagged share of non-farm income			0.156			-0.237
			(0.106)			(0.193)
Share of village with year-round passable road			-0.052			0.680***
			(0.038)			(0.215)
Share of village with daily market			0.036			0.180
			(0.032)			(0.194)
Distance from village to the nearest town			-0.015			-0.094

			(0.012)			(0.069)
Constant	2.101***	6.704***	6.835***	1.655***	1.861***	7.207***
	(0.369)	(0.469)	(0.524)	(0.390)	(0.422)	(0.924)
Year FE	No	Yes	Yes			
Province FE	Yes	Yes	Yes			
Observations	567	567	565	567	567	565
R2/ Chi2	0.0767	0.420	0.289	2298	3926	4210
Number of provinces	63	63	63	63	63	63
Arellano-Bond test for AR(1)				-7.365	-7.437	-5.325
Arellano-Bond test for AR(2)				2.939	2.837	1.197
Sargan test				311.1	310.8	62.81

Note: i) Authors' estimation from VHLSSs using province FE model (columns 1 to 3) and GMM model (columns 4 to 6), ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01 ** p<0.05, * p<0.1.

Table A.5: Factors related to provincial dissimilarity index

Explanatory variables	Dissimilarity Index			
	Model 1	Model 2	Model 3	Model 4
	(1)	(2)	(3)	(4)
Log of per capita expenditure	0.005***	-0.018***	-0.014**	-0.012**
	(0.001)	(0.006)	(0.006)	(0.006)
Gini index	0.029	0.066***	0.052***	
	(0.022)	(0.022)	(0.018)	
Log of state spending			-0.000	0.000
			(0.000)	(0.000)
Log of investment spending			-0.000	-0.000
			(0.001)	(0.001)
Log of population density			0.030***	0.030***
			(0.007)	(0.007)
Share of urban population			-0.000***	-0.000***
			(0.000)	(0.000)
Share of population with high-school diploma			-0.017	-0.019
			(0.020)	(0.021)
Share of ethnic minority population			0.000	0.000
			(0.000)	(0.000)
Share of wage income			0.016**	0.021***
			(0.007)	(0.007)
Share of non-farm income			0.015**	0.021**
			(0.007)	(0.008)
Constant	-0.040***	0.129***	0.146***	0.147***
	(0.096)	(0.419)	(0.350)	(0.024)
Year FE	No	Yes	Yes	No
Province FE	Yes	Yes	Yes	Yes
Observations	630	630	630	630
Number of provinces	63	63	63	63
R2	0.112	0.0189	0.0339	0.0226
sigma_u	0.0118	0.0137	0.0264	0.0286
sigma_e	0.00573	0.00508	0.00461	0.00471
rho	0.810	0.880	0.970	0.974

Notes: i) Authors' estimation from VHLSSs using province FE models, ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01, ** p<0.05, * p<0.1.

Table A.6: Relation between income growth, poverty, and income inequality

Explanatory variables	Dependent variable is the growth of per capita income					
	FE			GMM		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged log of per capita income	-0.169***	-0.589***	-0.579***	-0.156***	-0.119	-0.456***
	(0.025)	(0.050)	(0.055)	(0.037)	(0.082)	(0.091)
Lagged Gini index	-0.298	-0.258	-0.224	-0.347**	-0.607**	-0.607***
	(0.222)	(0.180)	(0.167)	(0.170)	(0.270)	(0.218)
Lagged poverty gap	-0.236***	-0.032	0.027	-0.240**	-0.091	-0.217
	(0.070)	(0.096)	(0.099)	(0.094)	(0.152)	(0.210)
Lagged log of other State spending			-0.002			-0.003
			(0.002)			(0.002)
Lagged log of investment spending			0.005			0.004
			(0.007)			(0.009)
Lagged log of population density			-0.279***			0.054***
			(0.072)			(0.013)
Lagged share of urban population			0.001			0.003***
			(0.001)			(0.001)
Lagged share of population with high-school diploma			-0.124			-0.024
			(0.317)			(0.305)
Lagged share of ethnic minority population			0.001			-0.000
			(0.000)			(0.000)

	(1)	(2)	(3)	(4)	(5)	(6)
Lagged share of wage income			0.188			0.083
			(0.117)			(0.123)
Lagged share of non-farm income			-0.060			-0.212
			(0.134)			(0.165)
Constant	1.809***	5.143***	4.536***	1.721***	1.471**	4.306***
	(0.272)	(0.432)	(0.489)	(0.348)	(0.746)	(0.789)
Year FE	No	Yes	Yes			
Province FE	Yes	Yes	Yes			
Observations	567	567	567	567	567	567
R2/ Chi2	0.137	0.485	0.515	2390	2632	2023
Number of provinces	63	63	63	63	63	63
Arellano-Bond test for AR(1)				-5.118	-4.641	-5.017
Arellano-Bond test for AR(2)				2.257	1.338	0.710
Sargan test				144.9	82.10	74.92

Note: i) Authors' estimation from VHLSSs using province FE model (columns 1 to 3) and GMM model (columns 4 to 6), ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01, ** p<0.05, * p<0.1.

Table A.7: Relation between expenditure growth, poverty gap, and inequality

Explanatory variables	Dependent variable is the growth of per capita expenditure (Log Y_t – Log Y_{t-1})					
	FE			GMM		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged log of per capita expenditure	-0.123***	-0.813***	-0.909***	-0.119***	-0.252**	-0.856***
	(0.009)	(0.062)	(0.065)	(0.013)	(0.102)	(0.134)
Lagged Gini index	-0.531	0.141	0.210	-0.711**	-0.573	-0.103
	(0.351)	(0.149)	(0.160)	(0.338)	(0.395)	(0.276)
Lagged poverty gap	-0.080	-0.165	-0.638***	0.105	-0.351	-2.192***
	(0.260)	(0.183)	(0.209)	(0.254)	(0.611)	(0.554)
Lagged log of other State spending			0.002**			0.002
			(0.001)			(0.002)
Lagged log of investment spending			0.003			0.011
			(0.005)			(0.008)
Lagged log of population density			-0.092			0.039**
			(0.068)			(0.015)
Lagged share of urban population			-0.001			0.004***
			(0.001)			(0.001)
Lagged share of population with high-school diploma			0.030			0.018
			(0.266)			(0.178)
Lagged share of ethnic minority population			0.002***			0.001*
			(0.000)			(0.000)
Lagged share of wage income			0.258***			0.379***
			(0.094)			(0.107)
Lagged share of non-farm income			0.144			0.184*
			(0.102)			(0.109)
Constant	1.396***	6.613***	7.100***	1.413***	2.367***	6.864***
	(0.124)	(0.494)	(0.507)	(0.076)	(0.773)	(1.008)
Year FE	No	Yes	Yes			
Province FE	Yes	Yes	Yes			
Observations	567	567	567	567	567	567
R2/ Chi2	0.091	0.894	0.904	2234	4069	7018
Number of provinces	63	63	63	63	63	63
Arellano-Bond test for AR(1)				-7.365	-6.458	-6.183
Arellano-Bond test for AR(2)				2.939	4.202	3.671
Sargan test				311.1	136.2	115.4

Note: i) Authors' estimation from VHLSSs using province FE model (columns 1 to 3) and GMM model (columns 4 to 6), ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01, ** p<0.05, * p<0.1.

Table A.8: Relation between expenditure growth, poverty severity, and inequality

Explanatory variables	Dependent variable is the growth of per capita expenditure (Log Y_t – Log Y_{t-1})					
	FE			GMM		
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6
	(1)	(2)	(3)	(4)	(5)	(6)
Lagged log of per capita expenditure	-0.116***	-0.804***	-0.891***	-0.113***	-0.241***	-0.782***
	(0.008)	(0.060)	(0.062)	(0.012)	(0.092)	(0.128)
Lagged Gini index	-0.592*	0.131	0.180	-0.855**	-0.605	-0.263
	(0.352)	(0.148)	(0.160)	(0.353)	(0.390)	(0.269)
Lagged poverty severity	0.387	-0.194	-1.002***	0.522	-0.600	-3.668***
	(0.516)	(0.314)	(0.329)	(0.575)	(1.196)	(1.093)
Lagged log of other State spending			0.002**			0.002
			(0.001)			(0.002)
Lagged log of investment spending			0.003			0.010
			(0.005)			(0.009)
Lagged log of population density			-0.099			0.039***
			(0.068)			(0.015)
Lagged share of urban population			-0.001			0.004***
			(0.001)			(0.001)
Lagged share of population with high-school diploma			0.034			-0.019
			(0.265)			(0.175)
Lagged share of ethnic minority population			0.002***			0.001
			(0.000)			(0.000)
Lagged share of wage income			0.270***			0.399***
			(0.093)			(0.106)
Lagged share of non-farm income			0.150			0.207*
			(0.102)			(0.109)
Constant	1.343***	6.535***	6.925***	1.399***	2.276***	6.260***
	(0.112)	(0.477)	(0.468)	(0.072)	(0.682)	(0.948)
Year FE	No	Yes	Yes			
Province FE	Yes	Yes	Yes			
Observations	567	567	567	567	567	567
R2/ Chi2	0.0744	0.420	0.246	1883	4033	6803

Number of provinces	63	63	63	63	63	63
Arellano-Bond test for AR(1)				-7.372	-6.495	-6.190
Arellano-Bond test for AR(2)				2.990	4.226	3.693
Sargan test				310.5	135.8	114.9

Note: i) Authors' estimation from VHLSSs using province FE model (columns 1 to 3) and GMM model (columns 4 to 6), ii) Robust standard errors in parentheses clustered at the province level, iii)*** p<0.01, ** p<0.05, * p<0.1.

Table A.9: Relation between expenditure growth, poverty, and inequality with interactions

Explanatory variables	Dependent variable is the growth of per capita expenditure (Log Y_t – Log Y_{t-1})							
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Lagged log of per capita expenditure	-0.659***	-0.767***	-0.491***	-0.525***	-0.505***	-0.486***	-0.519***	-0.505***
	(0.180)	(0.179)	(0.074)	(0.074)	(0.072)	(0.069)	(0.088)	(0.071)
Lagged log of per capita expenditure * Lagged log of other State spending	0.011							
	(0.009)							
Lagged log of per capita expenditure * Lagged log of investment spending		0.016*						
		(0.009)						
Lagged log of per capita expenditure * Lagged log of population density			0.001					
			(0.008)					
Lagged log of per capita expenditure * Lagged share of urban population				0.000				
				(0.000)				
Lagged log of per capita expenditure * Lagged share of population with high-school diploma					0.461*			
					(0.252)			
Lagged log of per capita expenditure * Lagged share of ethnic minority population						0.001*		
						(0.000)		
Lagged log of per capita expenditure * Lagged Gini index							0.114	
							(0.261)	
Lagged log of per capita expenditure * Lagged poverty rate								-0.188**
								(0.077)
Lagged log of other State spending	-0.097	0.002	0.000	0.001	0.001	0.001	0.001	0.001
	(0.083)	(0.002)	(0.002)	(0.002)	(0.001)	(0.002)	(0.002)	(0.002)
Lagged log of investment spending	0.003	-0.129	0.003	0.000	0.001	0.000	0.001	-0.002
	(0.007)	(0.082)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)	(0.007)
Lagged log of population density	0.035***	0.035***	0.030	0.039***	0.034***	0.030***	0.033***	0.031***

Explanatory variables	Dependent variable is the growth of per capita expenditure (Log Y_t – Log Y_{t-1})							
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
	(0.010)	(0.011)	(0.073)	(0.011)	(0.011)	(0.010)	(0.011)	(0.010)
Lagged share of urban population	0.002***	0.002***	0.002***	-0.002	0.002***	0.003***	0.002***	0.002***
	(0.001)	(0.001)	(0.001)	(0.005)	(0.001)	(0.001)	(0.001)	(0.001)
Lagged share of population with high-school diploma	0.175	0.163	0.063	0.203	-3.784*	0.102	0.145	0.177
	(0.156)	(0.155)	(0.138)	(0.146)	(2.155)	(0.135)	(0.146)	(0.159)
Lagged share of ethnic minority population	0.000	-0.000	-0.000	0.000	0.000	-0.006**	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.003)	(0.000)	(0.000)
Lagged Gini index	-0.887***	-0.902***	-0.873***	-0.882***	-0.934***	-1.008***	-1.924	-0.789***
	(0.133)	(0.148)	(0.155)	(0.149)	(0.143)	(0.156)	(2.299)	(0.175)
Lagged poverty rate	-0.135**	-0.183**	-0.117	-0.145**	-0.127*	-0.073	-0.116*	1.312**
	(0.062)	(0.075)	(0.077)	(0.071)	(0.067)	(0.071)	(0.069)	(0.600)
Lagged share of wage income	0.280***	0.302***	0.333***	0.291***	0.267***	0.326***	0.300***	0.353***
	(0.098)	(0.100)	(0.097)	(0.099)	(0.095)	(0.098)	(0.096)	(0.101)
Lagged share of non-farm income	0.217**	0.203**	0.141	0.151	0.161*	0.147	0.141	0.213**
	(0.097)	(0.091)	(0.097)	(0.094)	(0.086)	(0.096)	(0.090)	(0.092)
Constant	5.718***	6.568***	4.252***	4.582***	4.439***	4.257***	4.547***	4.416***
	(1.534)	(1.522)	(0.585)	(0.610)	(0.567)	(0.536)	(0.772)	(0.550)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	567	567	567	567	567	567	567	567
R2/ Chi2	6983	7747	8511	9686	7710	8503	7664	6568
Number of provinces	63	63	63	63	63	63	63	63
Arellano-Bond test for AR(1)	-6.628	-6.502	-6.668	-6.661	-6.734	-6.800	-6.654	-6.715
Arellano-Bond test for AR(2)	4.146	4.120	4.194	4.309	4.235	4.299	4.293	4.109
Sargan test	198.4	186.1	225.8	214	210.9	209.5	222.8	219.9

Note: i) Authors' estimation from VHLSSs using province GMM models, ii) Robust standard errors in parentheses clustered at the province level, iii)*** $p<0.01$, ** $p<0.05$, * $p<0.1$.

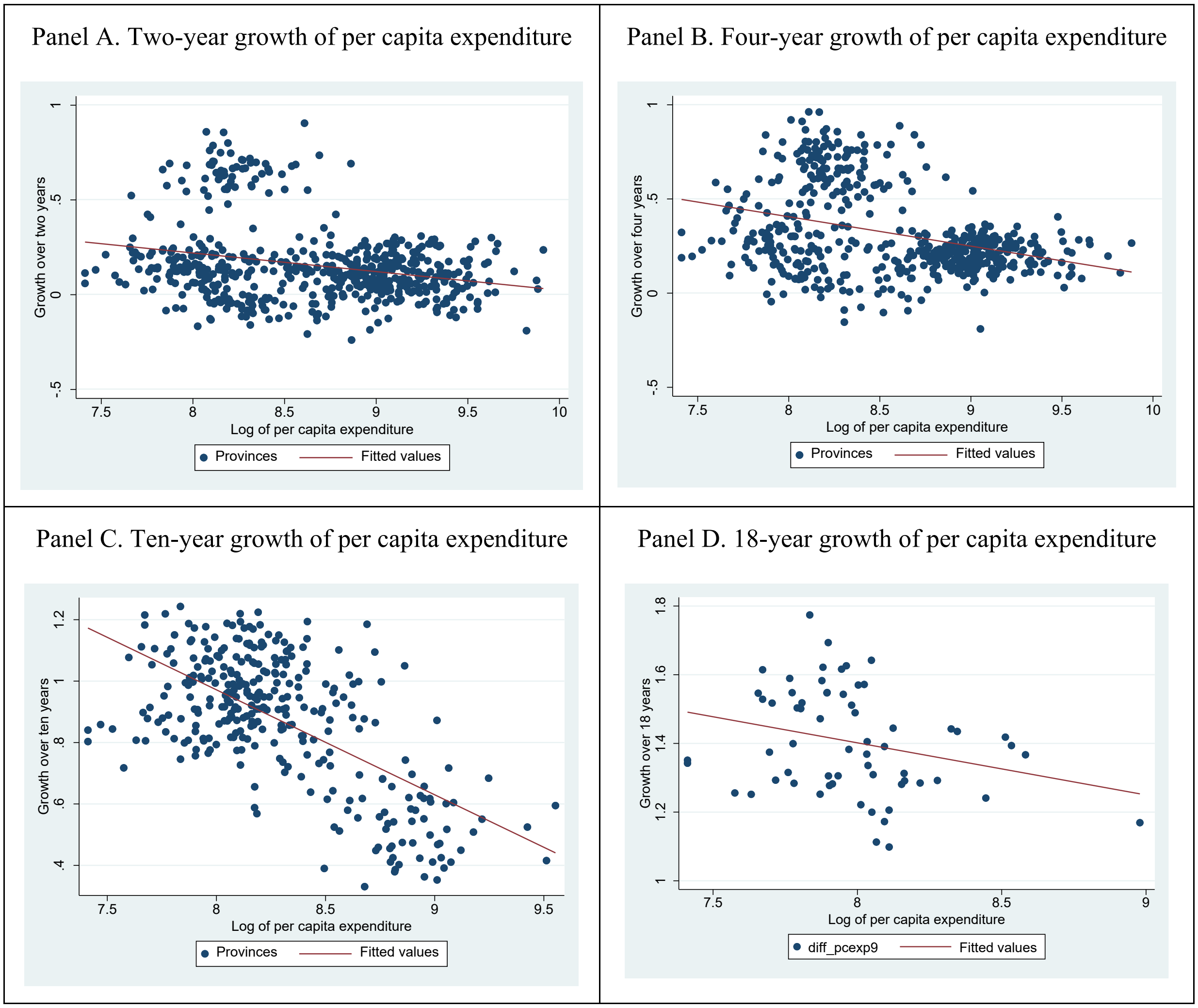
Table A.10: Relation between income growth, poverty, and inequality with interactions

Explanatory variables	Dependent variable is the growth of per capita income (Log Y_t – Log Y_{t-1})							
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
Lagged log of per capita income	-0.399***	-0.179	-0.312***	-0.254***	-0.339***	-0.378***	-0.355***	-0.326***
	(0.099)	(0.273)	(0.063)	(0.052)	(0.091)	(0.068)	(0.062)	(0.057)
Lagged log of per capita income * Lagged log of other State spending	0.004							
	(0.007)							
Lagged log of per capita income * Lagged log of investment spending		-0.009						
		(0.016)						
Lagged log of per capita income * Lagged log of population density			-0.008					
			(0.012)					
Lagged log of per capita income * Lagged share of urban population				-0.003***				
				(0.001)				
Lagged log of per capita income * Lagged share of population with high-school diploma					-0.339			
					(0.381)			
Lagged log of per capita income * Lagged share of ethnic minority population						-0.000		
						(0.000)		
Lagged log of per capita income * Lagged Gini index							-0.102***	
							(0.036)	
Lagged log of per capita income * Lagged poverty rate								-0.038**
								(0.017)
Lagged log of other State spending	-0.043	-0.004*	-0.005**	-0.004*	-0.005**	-0.004	-0.003	-0.005*
	(0.067)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Lagged log of investment spending	0.006	0.089	0.005	0.003	0.004	0.006	0.000	-0.000
	(0.008)	(0.145)	(0.007)	(0.007)	(0.008)	(0.008)	(0.007)	(0.007)
Lagged log of population density	0.042***	0.043***	0.110	0.062***	0.049***	0.044***	0.034***	0.042***

Explanatory variables	Dependent variable is the growth of per capita income (Log Y_t – Log Y_{t-1})							
	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8
	(0.010)	(0.009)	(0.112)	(0.012)	(0.011)	(0.010)	(0.012)	(0.009)
Lagged share of urban population	0.002***	0.002***	0.002***	0.034***	0.002***	0.002***	0.003***	0.002***
	(0.001)	(0.001)	(0.001)	(0.009)	(0.001)	(0.001)	(0.001)	(0.001)
Lagged share of population with high-school diploma	0.111	0.100	0.105	0.067	3.171	0.108	0.071	0.152
	(0.192)	(0.176)	(0.188)	(0.177)	(3.373)	(0.205)	(0.166)	(0.213)
Lagged share of ethnic minority population	-0.001**	-0.000	-0.001*	-0.000	-0.001**	0.000	-0.000	-0.000
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.004)	(0.000)	(0.000)
Lagged Gini index	-0.409**	-0.390**	-0.419**	-0.574***	-0.460**	-0.432**	-0.192	-0.421**
	(0.188)	(0.171)	(0.201)	(0.122)	(0.183)	(0.202)	(0.208)	(0.208)
Lagged poverty rate	-0.074	-0.056	-0.047	0.091	-0.032	-0.117	-0.098	0.148
	(0.096)	(0.121)	(0.106)	(0.083)	(0.115)	(0.111)	(0.091)	(0.149)
Lagged share of wage income	0.091	0.116	0.095	0.034	0.079	0.088	0.209**	0.037
	(0.108)	(0.100)	(0.105)	(0.110)	(0.106)	(0.108)	(0.099)	(0.099)
Lagged share of non-farm income	-0.223*	-0.221	-0.258**	-0.260**	-0.202	-0.227*	-0.224	-0.254**
	(0.122)	(0.139)	(0.131)	(0.131)	(0.147)	(0.137)	(0.142)	(0.129)
Constant	3.750***	1.794	3.011***	2.487***	3.224***	3.544***	3.529***	3.273***
	(0.904)	(2.396)	(0.536)	(0.434)	(0.784)	(0.578)	(0.551)	(0.456)
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	567	567	567	567	567	567	567	567
R2/ Chi2	3446	3460	3666	3055	3255	3335	3043	2985
Number of provinces	63	63	63	63	63	63	63	63
Arellano-Bond test for AR(1)	-4.913	-4.915	-4.998	-5.048	-5.176	-5.363	-4.898	-5.006
Arellano-Bond test for AR(2)	1.172	1.243	1.234	1.131	1.193	1.031	1.402	1.503
Sargan test	173.5	175.8	195.9	151.5	161.6	147.5	174.1	194.9

Note: i) Authors' estimation from VHLSSs using province GMM models, ii) Robust standard errors in parentheses clustered at the province level, iii)*** $p<0.01$, ** $p<0.05$, * $p<0.1$.

Figure A.1: Growth of per capita expenditure and initial expenditure level



Note: i) Authors' estimation using VHLSSs data

Appendix B: Further data description

The Vietnam Household Living Standard Surveys (VHLSS) are conducted biennially, covering approximately 45,000 households in each survey round. The VHLSSs are designed to be representative at the provincial level (63 provinces). Within the VHLSSs, a sub-sample of around 9,000 households is selected to collect expenditure data. However, this smaller sample is only representative at the regional level (6 regions). To obtain estimates that are representative at the provincial level, the full sample of 45,000 households should be utilized. Thus, we estimate per capita expenditure of households in the sample of 36,000 households using the "poverty mapping" imputation method (Elbers *et al.,* 2003). Estimating per capita expenditure consists of two steps. First, we estimate an expenditure model using the small-sample VHLSs (9,000 households). The dependent variable is the per capita expenditure, and the explanatory variables consist of household characteristics including demographics, ethnicity, education of household heads and household members, durables, housing conditions, and region dummies. We estimate separate models for urban and rural areas. The variables are selected using stepwise regressions. Only variables that are statistically significant at the 1% level and demonstrate reasonable signs are used in the final models. Second, we apply this expenditure model to the sample of 36,000 households (using the same variables that were employed in the expenditure model based on the small-sample VHLSSs) and predict per capita expenditure for these households. As a result, we have per capita expenditure data for the full sample of 45,000 households, and we use this data to estimate the per capita expenditure of provinces.